ACE Cyber Preliminary Literature Map: Fall 2025

1. 1. Project Information

Student Name(s): Teaundra Thompson

Project Title: IDS Resiliency Against Advanced Evasion Attacks

Date / Version: 12/15/2025 v2.1

Mentor/Advisor: Professor Henderson

2. Research Focus

a. Research Question (draft): What is the percentage drop of detection accuracy of the baseline IDS when faced with evasion attacks?

b. Generate a list of detailed "Keywords" and "Key Concepts": Evasion attacks, IDS integrity, perturbation, cybersecurity, ML models, anomaly detection, NIDS, MLP,SQL injection, Neural Networks, AI-based IDS, Evasion Mitigation, DDoS
(Example: phishing, human factors, awareness training, cyber resilience, social engineering)

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3. Supporting Literature Table

Citation (APA/IEEE)	Research Type (Empirical, Review, etc.)	Key Findings	Relevance to Research Question	Gap/ Limitations Identified	Empirical Study	Summary in 2–3 sentences	Why it matters to your project	What's missing / unresolved
Chio, C., & Freeman, D. (2018) . *Machine Learning and Security: Protecting Systems with Data and Algorithms*. O'Reilly Media, Inc.	Mixed method: Review, and empirical	-Details regarding evasion attacks and how their effect -Types of training and datasets -evasion attacks effectiveness against various ML models -Binary classifier evasion attack -Types of IDS -Cybersecurity in general	Very relevant; there is a section dedicated to evasion attacks against anomaly detection systems	-Gives an empirical review of a MLP IDS model facing attacks, but they were poisoning attacks not evasion.	No	This book has a plethora of information regarding ML/AI models, and also datasets. There is a dedicated section for evasion attacks against AI-based IDS	This source matters to my project because it covers multiple topics thoroughly and will be a good source to look at to garner ideas. This book also helped me to decide between poisoning and evasion techniques.	General overview of cybersecurity systems, so I need additional sources to cover the specifics of my research.

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Mohammed, D. F. (2024). *A Study of Network Intrusion Detection Systems Using Artificial Intelligence*. *12*(5).	Review	-Has details regarding incorporating AI into NIDs -Benefits of AI in cyber security -Has details regarding signature and anomaly-based detection	Rather relevant; focusses on a specific type of IDS and talks about the basics. Also helps to influence my project design.	No tangible data (graphs, tables, etc.)	No	This article has content regarding NIDs and it details the detection techniques that are beneficial. It briefly covers other topics related to training techniques and benchmarks.	This source matters to my project because it speaks about NIDS, and my IDS will be closely similar to a NIDS. This source can be used to develop my talking points.	This source does not have visual data to glean from. I will not be able to use this source to assist with benchmarking my IDS' performance
Adli, T. B., Amokrane, S.-B. B., Pavlović, B. Z., Laidouni, M. Z. M., & Benyahia, T. A. A. (2023). Anomaly network intrusion detection system based on NetFlow using machine/deep learning. *Military Technical Courier / Vojnotehnicki Glasnik*, *71*(4), 941–969. https://doi.org/10.5937/vojtehg71-46058	Hypothetico-deductive	-Anomaly based NIDS' and NetFlow analysis will benefit military networks -Mentions network protocols that can be used in conjunction with a NIDS	-Very relevant; it focuses on NIDs, which is an IDS model I am interested in. Discusses NetFlow, which makes data collection and processing easier than with pcaps.	Will not be able to conduct experiments using multiple neural network models.	Yes	This article has details regarding major IDS types but focuses on NIDSs. There is copious data and stats from various ML models and their metrics.	This article matters to my project because I can use it for reference when I am benchmarking my IDS. I also get an idea of what NetFlow is.	Does not specifically mention evasion attacks.

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Building an intrusion detection system on UNSW-NB15: Reducing the margin of error to deal with data overlap and imbalance. (n.d.). https://doi.org/10.1002/cpe.8242	Hypothetico-deductive	-Focuses on dataset UNSW-NB15 -In depth data on preprocessing and classification training -Different algorithms and their metrics on the dataset	Mildly relevant; focuses on one dataset and how to improve IDS decision boundaries on that dataset.	Focuses on a singular dataset, and I may not utilize this specific dataset. Does not simulate attacks.	Yes	This article topic is the resulting IDS from being trained on the UNSW-NB15 dataset. Has thorough data resulting from the experiment.	This source matters to my project because it gives some insight into the UNSW-NB15 data set and algorithms accuracy on said dataset. This data-set features modern attack behaviors unlike older datasets.	-Other datasets
M.s, M. S. R. S. R. (2025). The Rise of AI-Driven Network IDS: Innovations, Challenges, and Future Directions. *International Journal of AI, BigData, Computational and Management Studies*, *6*(1), 1–9. https://doi.org/10.63282/3050-9416.IJAIBDCMS-V6I1P101	Applied experimentation	-Benefits of AI-based IDS -Supervised and unsupervised learning -Challenges associated with AI-based IDS -ML models and techniques	Relevant; details about basic AI and ML models. Gives a couple examples of evasion techniques. Material to use for my conclusory section in my report.	Speaks about datasets in general; no specific dataset mentioned	Yes, partially	This article provides a general overview of AI-based IDSs and learning types. Provides data from a case study of a Generative Adversarial Network.	This source matters to my project because it will be useful when creating my report because it has information regarding the benefits of AI IDSs and how to remediate it's vulnerabilities.	Case study from a Decision tree model

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Ou, Y., Lin, Y., Zhang, Y., & Ou, Y. (2010). The Design and Implementation of Host-Based Intrusion Detection System. *2010 Third International Symposium on Intelligent Information Technology and Security Informatics*, 595–598. https://doi.org/10.1109/IITSI.2010.127	Empirical	- Hybrid HIDS using log file analysis and back propagation (BP) neural network analysis algorithm -BP neural network is multilayer and minimizes error by back-propagation	Relevant; gives good insight into creating a baseline HIDS. Gives thorough info on log file analysis.	Does not address evasion attacks specifically	Yes	The authors created a HIDS using log analysis and a BP neural network for anomaly detection. They argue that the two methods combined create a robust HIDS.	This source matters to my project because it provides inspiration for creating an HIDS that utilizes log file analysis,	How was the HIDS deployed? On a vm with prepopulated log files?
Elam, M., Mink, D., Bagui, S. S., Plenkers, R., & Bagui, S. C. (2025). Introducing UWF-ZeekData24: An Enterprise MITRE ATT&CK Labeled Network Attack Traffic Dataset for Machine Learning/AI. *Data*, *10*(5), 59. https://doi.org/10.3390/data10050059	Applied Research	- Details about UFW-ZeekData24 dataset -The dataset follows MITRE ATT&CK Framework -Dataset is in a Zeek data format -Can be used for both training and testing	Relevant; information regarding a potential dataset. The dataset follows the MITRE ATT&CK framework which makes it easy for classification	The dataset is heavily imbalance with one tactic making up 90% of malicious samples.	Yes	This article gives details about the UFW-ZeekData24 dataset; how it was created, and what kind of attacks it contains. The dataset is based on the MITRE ATT&CK framework making it intuitive, for classification.	This source provides information on a potential dataset that I may use.	Not clear how the Zeek logs were parsed to create the dataset

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Ganesan, A., & Sarac, K. (2021). Mitigating Evasion Attacks on Machine Learning based NIDS Systems in SDN. *2021 IEEE 7th International Conference on Network Softwarization (NetSoft)*, 268–272. https://doi.org/10.1109/NetSoft51509.2021.9492526	Hypothetico-deductive	-Evasion attacks and packet perturbation -Focuses on Network Intrusion Detection Systems -Mitigate evasion attacks by utilizing multiple ML models	Very relevant, this article provides ideas for evasion mitigation	The variables in the research paper may require significant computational power	Yes	This article creates and evaluates mitigation strategies for evasion attacks on a NIDS in a software defined network. Utilizes multiple ML models in the IDS to increase detection of evasion attacks.	I can compare my data with the data in this research article	n/a
Ayub, Md. A., Johnson, W. A., Talbert, D. A., & Siraj, A. (2020). Model Evasion Attack on Intrusion Detection Systems using Adversarial Machine Learning. *2020 54th Annual Conference on Information Sciences and Systems (CISS)*, 1–6. https://doi.org/10.1109/CISS48834.2020.1570617116	Hypothetico-deductive	-Uses a Multilayer Perceptron Network (MLP) for the IDS model and models how a Jacobian-based Saliency Map attack can evade detection. Trains model with two datasets.	Very relevant; the topic of the paper is very similar to my research topic	Training the IDS with two different datasets may be resource intensive.	Yes	The authors create a MLP IDS model with a high accuracy detection rate. Then they create adversarial attacks using JSMA with causes the detection rate to drop significantly.	Gives a good example of adversarial evasion and serves for inspiration.	n/a

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Bace, R., & Mell, P. (2001). NIST Special Publication on Intrusion Detection Systems. https://apps.dtic.mil/sti/html/tr/ADA393326/index.html	Technical Report	-Defines what an IDS is -Strengths and limitations of IDSs	Very relevant; provides a solid definition of what an IDS is and gives me guidelines to follow	Does not address adversarial evasion attacks; this article is 24 years old.	No	This report provides thorough information detailing what an IDS is and it's functions. It also provides details on different types of IDSs.	This source provides information on how my IDS should function and gives me information on its components.	Lacks information of ML IDSs, and how to deal with advanced evasion techniques.

5. Preliminary Themes & Gaps

Emerging Themes:

- The increasing sophistication of evasion attacks necessitates rapid agile and adaptive systems
- NIDs are the most robust version of an IDS
- Neural networks are best for detecting behavioral patterns in traffic.

-Utilizing multiple ML models in an IDS makes the system very robust and against evasion attacks

Potential Research Gap:

I need sources that mention deploying an ML algorithm in a cloud environment or on a vm.

6. Next Steps

What additional areas will you search/explore in the next 2 weeks?

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I will research deployment options in AWS to determine the feasibility and performance impact of using such a resource to host a ML model for anomaly detection/ I will also explore what frameworks and tools would be necessary for creating and implementing my ML model to determine what hosting solution would be best.

What gaps need more sources?

I need to do more research on different types of frameworks for ML models, and what would be most suitable for this project.

Any difficulties so far?

n/a.